



FINAL EXAM BRIDGING SEMESTER II, SESSION 2020/2021

COURSE CODE : IFC 1024
COURSE NAME : CHEMISTRY
PROGRAMME : BRIDGING SPACE
DURATION : 3 HOURS
DATE : 06 JULY 2021

INSTRUCTION TO CANDIDATES:

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.
2. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

This booklet consists of (21) printed pages including this page.

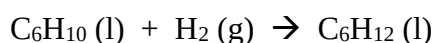
QUESTION 1 (20 MARKS)

- a) A 1.60 dm³ sample of a mixture of methane gas, CH₄ and oxygen gas, measured at 25 °C and 101 kPa, was allowed to react in a bomb calorimeter in which, had a heat capacity of 5.30 kJ/K altogether with its contents. The complete combustion of the methane gas to carbon dioxide gas and water caused temperature rise in the calorimeter of 6.28 K. Given that ΔH_c° (CH₄) is -560 kJ/mol.

- i. Define the standard enthalpy of combustion of methane
- ii. Write a thermochemical equation for the combustion of methane
- iii. Knowing the ΔH_c° of methane, calculate the number of mole of methane in the mixture.
- iv. Calculate the total moles of gases in the calorimeter.

(1+1+3+2 marks)

- b) Cyclohexene, C₆H₁₀ undergoes hydrogenation to produce cyclohexane, C₆H₁₂ according to the equation given below :



$$\Delta H_f^\circ (\text{C}_6\text{H}_{10}) = -30 \text{ kJ/mol}$$

$$\Delta H_f^\circ (\text{C}_6\text{H}_{12}) = -3930 \text{ kJ/mol}$$

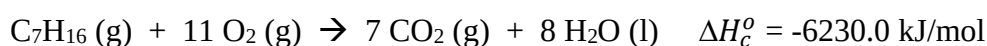
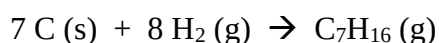
- i. Calculate the standard enthalpy of hydrogenation, ΔH° of cyclohexene.
- ii. Sketch the enthalpy diagram of this reaction

(2+2 marks)

- c) Propane is a hydrocarbon gas that is commonly used for cooking. The standard enthalpy of combustion of propane is -225.1 kJ/mol. 5 g of propane is used to heat 6 L of water. Assuming that no heat is lost to the surroundings, what is the increase in the water's temperature? Given that the specific heat capacity of water is $s(\text{H}_2\text{O}) = 4.18 \text{ J/g}^\circ\text{C}$ and density of water is 1 g/L.

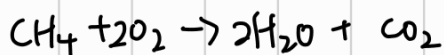
(5 marks)

- d) Given the following data, calculate the enthalpy change for the formation of 60 g of heptane, C₇H₁₆.



(4 marks)

1) (a) the heat absorbed when 1 mol of methane is burnt completely in excess oxygen to produce CO_2 and H_2O in standard condition



$$q = C_p \cdot \Delta T$$

$$= 5.30 \text{ kJ/K} \times 6.28$$

$$= 33.284 \text{ kJ}$$

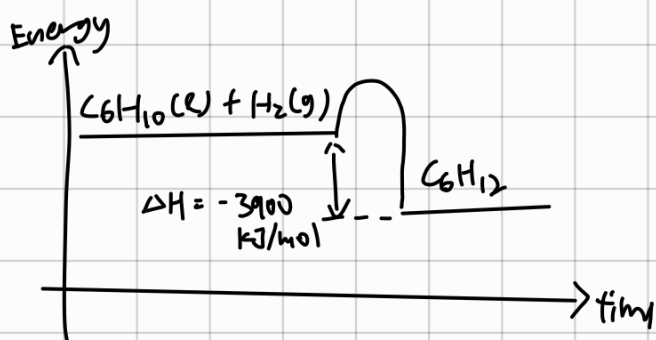
$$\Delta H = \frac{-q}{\text{mol}}$$

$$-560 \text{ kJ/mol} = \frac{-33.284 \text{ kJ}}{x}$$

$$x = 0.0594 \text{ mol of CH}_4$$

0.0594 mol of CH_4 will produce 0.0594 mol of CO_2

$$\begin{aligned} \Delta H_{\text{reaction}} &= -3930 \text{ kJ/mol} - (-30 \text{ kJ/mol}) \\ &= -3900 \text{ kJ/mol} \end{aligned}$$



(c) m.m of propyne, $\text{C}_3\text{H}_4 = 44$

$$\text{mol of C}_3\text{H}_4 = \frac{5}{44} = 0.11364 \text{ mol}$$

$$\Delta H = \frac{q}{\text{mol}}$$

$$-25.1 \text{ kJ/mol} = \frac{-q}{0.11364}$$

$$q = 25.58 \text{ kJ}$$

$$q_1 = q_{\text{water}}$$

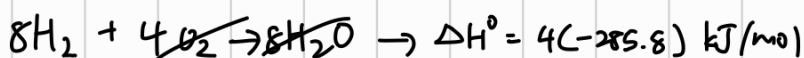
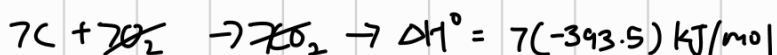
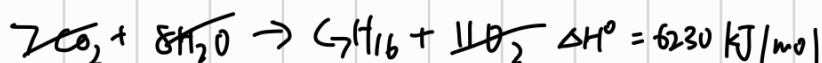
$$q_1 = m s \Delta T$$

$$25.58 \times 1000 = \left(\frac{11}{12} \times 6 \times \frac{1000}{12} \right) (4.18) (\Delta T)$$

$$25580 = 6000 (4.18) (\Delta T)$$

$$\Delta T = 1.02^\circ\text{C}$$

(d) mol of heptane = $\frac{60}{100} = 0.6 \text{ mol}$

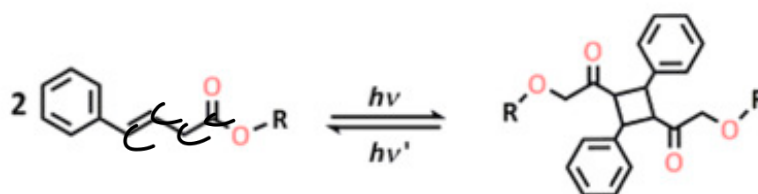


$$\Delta H = 2332.3 \text{ kJ/mol}$$

QUESTION 2 (10 MARKS)

- a) Photodimerization is a chemical process when one photoexcited molecule reacts with one unexcited molecule of the same species. One example of this process is the reaction of cinnamate as shown below.

Cinnamate



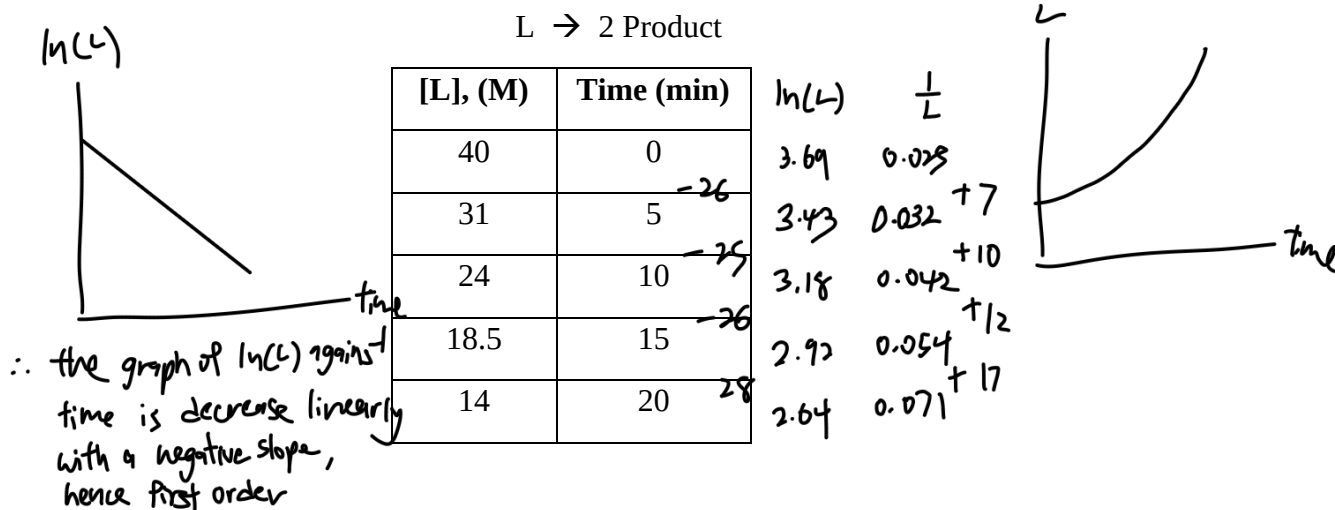
- This reaction is reported as a second order reaction. Write a rate law equation for the forward reaction. $R = k[A]^2$
- Based on your answer in a(i), calculate the initial concentration of cinnamate, $[\text{cinnamate}]_0$ when k is 3.85 L/mol.s and initial rate is found to be $7.4 \times 10^{-5} \text{ mol/L.s}$

$$7.4 \times 10^{-5} = 3.85 [A]^2$$

$$A = 0.00436 \text{ M}$$

(1+3 marks)

- b) The experimental data for elimination reaction is shown below :



Determine the rate law equation for the reaction using graphical method. Draw your graphs and show all your assumptions.

$$R = k[L]$$

(6 marks)

$$\text{H}_2\text{SO}_4 \quad \text{mol H} = \frac{2}{2+1+4} = \frac{2}{7}$$

QUESTION 3 (10 MARKS)

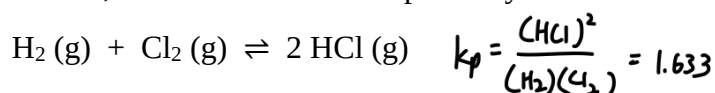
673 k

- a) 1 mole of N_2 and 3 mole of H_2 are placed in a flask at $400^\circ C$. Calculate the total pressure of the system at initial, P_0 in the function of x (the change) if the mole fraction of NH_3 is 0.43 at equilibrium. $P = [x]RT$

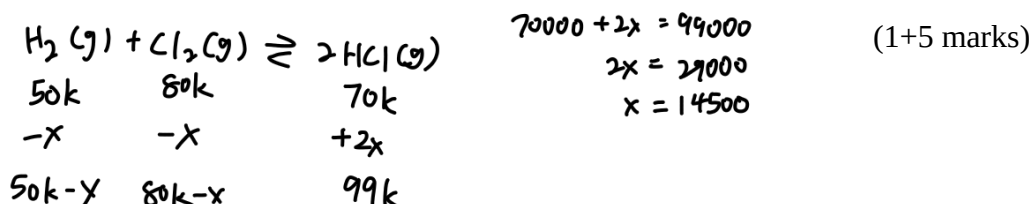
$$P_i = x_i P_{total}$$

$$27x^4 - 105x^3 + 162x^2 - 108x + 27 \quad (4 \text{ marks})$$

- b) Hydrochloric acid, HCl is produced from the reaction between hydrogen gas and chlorine gas. At temperature $T^\circ C$, the partial pressures of H_2 , Cl_2 and HCl in a 1 dm^3 equilibrium mixture are 50 kPa, 60 kPa and 70 kPa respectively.



- Calculate the value of equilibrium constant, K_p .
- When additional Cl_2 is added, its new partial pressure is 80 kPa. The new equilibrium partial pressure of HCl is 99 kPa. Determine the partial pressure of H_2 and Cl_2 in the new equilibrium mixture if temperature remains constant.



$$\begin{aligned}
 P(H_2) &= 50000 - 14500 = 35500 \text{ Pa} \\
 P(Cl_2) &= 80000 - 14500 = 65500
 \end{aligned}$$

 NH_3

$$3) \text{ a) } P_{H_2} = \frac{3}{1+3} (P_i) = 0.75 P_i$$

$$P_{H_2} = \frac{1}{1+3} (P_i) = 0.25 P_i$$

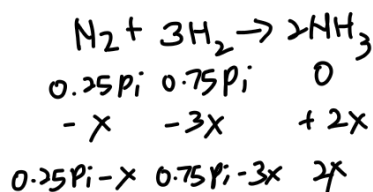
$$P_{NH_3} = 2x$$

$$0.43 = \frac{P_{NH_3}}{P_{H_2} + P_{H_2} + P_{NH_3}}$$

$$0.43 = \frac{2x}{P_i - 2x}$$

$$0.43 P_i - 0.86x = 2x$$

$$P_i = 6.65x$$



$$\begin{aligned}
 \text{Total initial pressure} &= 0.25(6.65x) + 0.75(6.65x) \\
 &= 6.65x
 \end{aligned}$$

QUESTION 4 (10 MARKS)

- a) Calculate the base dissociation constant, K_b , of sodium hydrosulfide, NaHS given the acid dissociation constant, K_a , of hydrogen sulfide is 3.5×10^{-6} .

$$K_w = K_a \times K_b \quad K_b = 2.86 \times 10^{-9} \quad (2 \text{ marks})$$

$$1 \times 10^{-14} = 3.5 \times 10^{-6} \times K_b$$

- b) Pyridine, C_5H_5N , is a weak base and combine with X to form a buffer solution.

i. Give an example of X. C_5H_6NCl

- ii. Calculate the amount (mole) of X to be used to be added to a beaker containing 120 mL of 1.1 M C_5H_5N to form a buffer with a pH of 6.1. Given the base dissociation constant, K_b of C_5H_5N is 1.7×10^{-9} .

$$pOH = -\log K_b + \log \left(\frac{C_5H_6NCl}{C_5H_5N} \right) \quad \text{mol of } C_5H_5N = 0.12 \times 1.1 = 0.132 \text{ mol} \quad (1+3 \text{ marks})$$

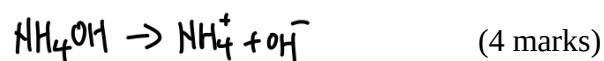
$$7.9 = -\log(1.7 \times 10^{-9}) + \log \left(\frac{x}{0.132} \right), \quad x = 0.0178 \text{ mol}$$

- c) Given that only 11.2 % of 1.0 M ammonium hydroxide, NH_4OH , ionizes in water. Calculate the base dissociation constant, K_b , of the ammonium hydroxide.

$$\% \text{ ionization} = \frac{\text{mol ionized}}{\text{mol available}} \times 100$$

$$11.2 = \frac{x}{1.0} \times 100$$

$$x = 0.112 \text{ M}$$



$$\begin{array}{ccc} 1 & 0 & 0 \\ -x & +x & +x \\ 1-x & x & x \end{array}$$

$$K_b = \frac{0.112^2}{1-0.112}$$

$$= 0.014$$

$$(a) K_a \times K_b = K_w$$

$$3.5 \times 10^{-6} \times K_b = 1 \times 10^{-14}$$

$$K_b = 2.857 \times 10^{-9}$$

$$(b) (i) \text{ base : } C_5H_5N$$

$$\text{conjugate acid : } C_5H_6N^+$$

$$\text{salt of conjugate acid : } C_5H_6NCl$$

$$(ii) pOH = pK_b + \log\left(\frac{[C_{acid}]}{[base]}\right)$$

$$14 - 6.1 = -\log(1.7 \times 10^{-9}) + \log\left(\frac{[C_{acid}]}{0.12 \times 1.1}\right)$$

$$7.9 = 8.77 + \log\left(\frac{[C_{acid}]}{0.132}\right)$$

$$\log\left(\frac{x}{0.132}\right) = -0.87$$

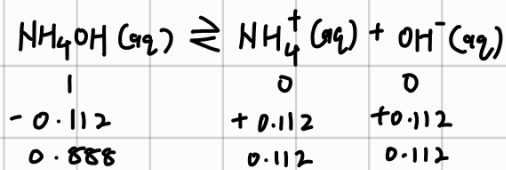
$$\frac{x}{0.132} = 10^{-0.87}$$

$$x = 0.018 \text{ mol}$$

$$[C_5H_6N^+] = [x] = [C_5H_6NCl] = 0.018 \text{ mol}$$

$$(c) \% \text{ ionisation} = 11.2 = \frac{\text{mol ionized}}{\text{mol available}} \times 100$$

$$\text{mol ionized} = 0.112 \text{ M}$$



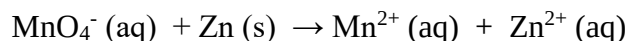
$$K_b = \frac{[NH_4^+][OH^-]}{[NH_4OH]}$$

$$= \frac{0.112^2}{0.888}$$

$$= 0.014$$

QUESTION 5 (20 MARKS)

a) Given the redox reaction equation below:



- Complete and balance the above equation in acidic solution.
- Calculate the standard potential produced from the above redox reaction.
- Write the Q_c expression for the above reaction

(5+2+1 marks)

b) Consider a concentration cell (voltaic cell) at temperature of 25°C. The setup consists of chromium electrodes on both sides. One side is filled with 300 mL of 0.2 M chromium sulfate, $\text{Cr}_2(\text{SO}_4)_3$ solution and the other side is 300 mL of 3.6 M $\text{Cr}_2(\text{SO}_4)_3$ solution.

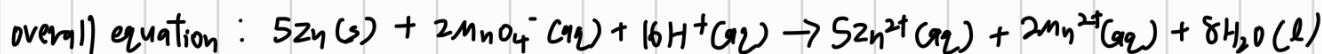
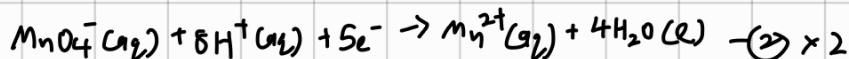
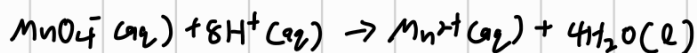
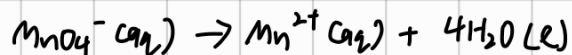
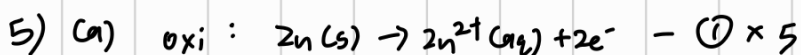
- Identify which concentration of chromate solution should be on the anode and cathode.
- Calculate the voltage produced by the cell.

(1+4 marks)

c) Consider 0.2 M of tin bromide, SnBr_2 in an electrolytic cell with graphite electrodes being used on both sides.

- Write out all possible chemical equations that may occur on both anode and cathode.
- Based on your answer in c(i), identify the major product that will be formed at the anode and cathode.
- If 0.2 A of current is applied on cell for 55 minutes, calculate the mass of product formed at the cathode.

(2+2+3 marks)

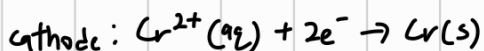
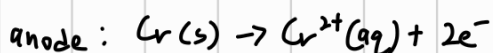


$$E^\circ_{\text{cell}} = E^\circ_{\text{cathode}} - E^\circ_{\text{anode}}$$

$$= (+1.51) - (-0.76)$$

$$= +2.37 \text{ V}$$

$$Q_c = \frac{[\text{Mn}^{2+}][\text{Zn}^{2+}]}{[\text{MnO}_4^-]}$$

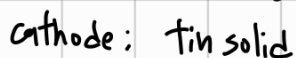
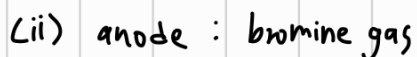
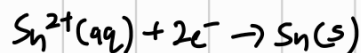
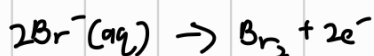
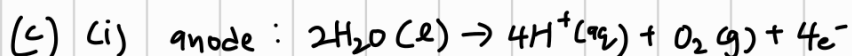


$$E^\circ_{\text{cell}} = 0 \text{ V}$$

$$E_{\text{cell}} = E^\circ_{\text{cell}} - \frac{0.0592}{n} \log Q$$

$$= 0 - \frac{0.0592}{2} \log \left(\frac{0.2}{3.6} \right)$$

$$= 0.037 \text{ V}$$



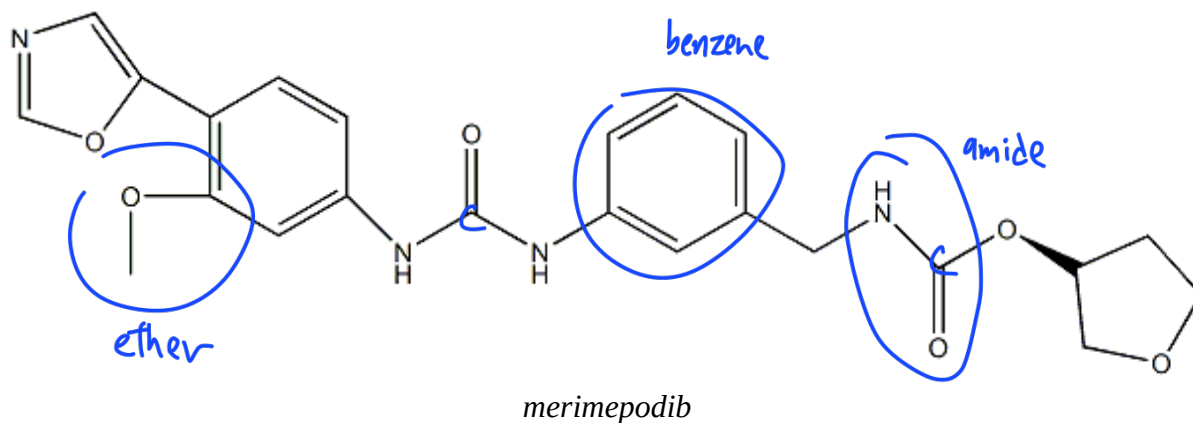
(iii) $m = \frac{AIt}{zF}$

$$= \frac{119(0.2)(55 \times 60)}{2 \times 96500}$$

$$= 0.407 \text{ g}$$

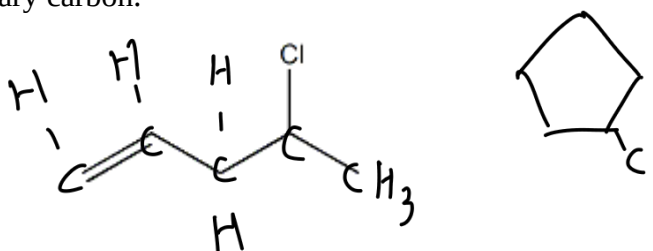
QUESTION 6 (10 MARKS)

- a) Image below shows a structural formula of merimepodib, an-antiviral drugs which has been clinically trialed for Covid-19 cure. Draw and list TWO (2) functional groups from the structure.



(2 marks)

- b) The following structure is one isomer of C_5H_9Cl . Draw ONE (1) isomer of C_5H_9Cl that does not contain any primary carbon.



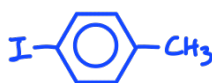
(2 marks)

- c) Draw the structure of the following molecules.

i. Trans-1,3-dichlorocyclopentane

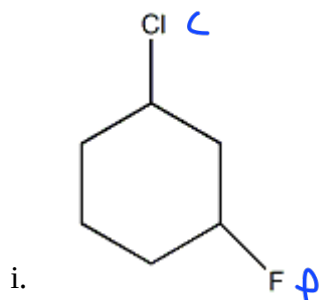


ii. p-iodotoluene



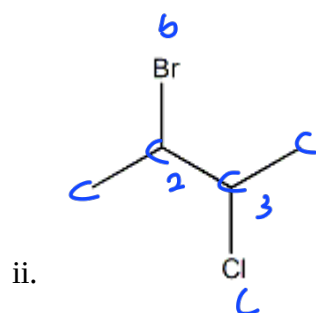
(1+1 marks)

- d) Write the IUPAC name for the following structures.



m-1-chloro-3-fluorocyclohexane

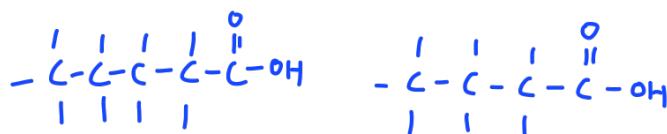
i.



2-bromo-3-chlorobutane

(1+1 marks)

e) Draw TWO (2) alcohol structures from the chemical formula $C_5H_{10}O_2$.



(2 marks)

The Periodic Table

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
IA	IIA	IIIA	IVA	VA	VIA	VIIA	VIIIA	VIIIA	VIIIA	VIIIA	VIIIA	VIIIA	VIIIA	VIIIA	VIIIA	VIIIA	VIIIA
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
H	He	Li	Be	B	C	N	O	F	Ne	Na	Mg	Al	Si	P	S	Cl	Ar
1.01	4.00	6.94	9.01	10.81	12.01	14.01	16.00	19.00	20.18	22.99	24.31	26.98	28.09	30.97	32.07	35.45	39.95
19	20	39.1	40.08	50.94	52.00	54.94	55.85	58.93	58.69	63.55	65.39	69.72	72.61	74.92	78.96	79.90	83.80
K	Ca	K	Ca	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
39.1	40.08	39.1	40.08	50.94	52.00	54.94	55.85	58.93	58.69	63.55	65.39	69.72	72.61	74.92	78.96	79.90	83.80
37	38	37	38	41	42	43	44	45	46	47	48	49	50	51	52	53	54
Rb	Sr	Rb	Sr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
85.47	87.62	85.47	87.62	92.91	95.94	(98)	101.07	102.91	106.42	107.87	112.41	114.82	118.71	121.76	127.6	126.9	131.29
55	56	55	56	73	74	75	76	77	78	79	80	81	82	83	84	85	86
Cs	Ba	Cs	Ba	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
132.9	137.3	132.9	137.3	180.9	183.9	186.2	190.2	192.2	195.1	197.0	200.6	204.4	207.2	209	(209)	(210)	(222)
87	88	87	88	105	106	107	108	109	110	111							
Fr	Ra	Fr	Ra	Db	Sg	Bh	Hs	Mt	Ds	Rg							
(223)	(226)	(223)	(226)	(262)	(263)	(264)	(265)	(268)	(271)	(272)							

List of Selected Constant Values

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

Unit and Conversion Factor

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

Standard Reduction Potential (E°_{SRP})

Half-Reaction	E°_{red}
$\text{K}^+ + \text{e}^- \rightleftharpoons \text{K}$	-2.924
$\text{Ba}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ba}$	-2.900
$\text{Ca}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ca}$	-2.760
$\text{Na}^+ + \text{e}^- \rightleftharpoons \text{Na}$	-2.712
$\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$	-2.375
$\text{H}_2 + 2\text{e}^- \rightleftharpoons 2\text{H}^-$	-2.230
$\text{Al}^{3+} + 3\text{e}^- \rightleftharpoons \text{Al}$	-1.706
$\text{Mn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mn}$	-1.040
$\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$	-0.763
$\text{Cr}^{3+} + 3\text{e}^- \rightleftharpoons \text{Cr}$	-0.740
$\text{S} + 2\text{e}^- \rightleftharpoons \text{S}^{2-}$	-0.508
$2\text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{C}_2\text{O}_4$	-0.490
$\text{Cr}^{3+} + \text{e}^- \rightleftharpoons \text{Cr}^{2+}$	-0.410
$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	-0.409
$\text{Co}^{2+} + 2\text{e}^- \rightleftharpoons \text{Co}$	-0.280
$\text{Ni}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ni}$	-0.230
$\text{Sn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Sn}$	-0.136
$\text{Pb}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pb}$	-0.126
$\text{Fe}^{3+} + 3\text{e}^- \rightleftharpoons \text{Fe}$	-0.036
$2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$	0.000
$\text{S}_4\text{O}_6^{2-} + 2\text{e}^- \rightleftharpoons 2\text{S}_2\text{O}_3^{2-}$	0.089
$\text{Sn}^{4+} + 2\text{e}^- \rightleftharpoons \text{Sn}^{2+}$	0.150
$\text{Cu}^{2+} + \text{e}^- \rightleftharpoons \text{Cu}^+$	0.158
$\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	0.340
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightleftharpoons 4\text{OH}^-$	0.401
$\text{Cu}^+ + \text{e}^- \rightleftharpoons \text{Cu}$	0.522
$\text{I}_3^- + 2\text{e}^- \rightleftharpoons 3\text{I}^-$	0.534
$\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightleftharpoons \text{MnO}_2 + 4\text{OH}^-$	0.588
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{O}_2$	0.682
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	0.770
$\text{Hg}_2^{2+} + 2\text{e}^- \rightleftharpoons \text{Hg}$	0.796
$\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$	0.800
$\text{Hg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Hg}$	0.851
$\text{H}_2\text{O}_2 + 2\text{e}^- \rightleftharpoons 2\text{OH}^-$	0.880
$\text{HNO}_3 + 3\text{H}^+ + 3\text{e}^- \rightleftharpoons \text{NO} + 2\text{H}_2\text{O}$	0.960
$\text{Br}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$	1.087
$2\text{IO}_3^- + 12\text{H}^+ + 10\text{e}^- \rightleftharpoons \text{I}_2 + 6\text{H}_2\text{O}$	1.190
$\text{CrO}_4^{2-} + 8\text{H}^+ + 3\text{e}^- \rightleftharpoons \text{Cr}^{3+} + 4\text{H}_2\text{O}$	1.195
$\text{Pt}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pt}$	1.200
$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{Mn}^{2+} + 2\text{H}_2\text{O}$	1.208
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	1.229
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.330
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	1.358
$\text{PbO}_2 + 4\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{Pb}^{2+} + 2\text{H}_2\text{O}$	1.467
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.491
$\text{Au}^+ + \text{e}^- \rightleftharpoons \text{Au}$	1.680
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	1.776
$\text{Co}^{3+} + \text{e}^- \rightleftharpoons \text{Co}^{2+}$	1.842
$\text{S}_2\text{O}_8^{2-} + 2\text{e}^- \rightleftharpoons 2\text{SO}_4^{2-}$	2.050
$\text{O}_3(\text{g}) + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{O}_2(\text{g}) + \text{H}_2\text{O}$	2.070
$\text{F}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons 2\text{HF}(\text{aq})$	3.030

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